

- 4 (a) A circuit is set up as shown in Fig. 4.1. Cell A has an electromotive force (e.m.f.) of 2.0 V and negligible internal resistance. XY is a uniform wire of length 100.0 cm and resistance 5.0 Ω .

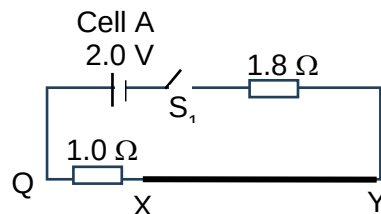


Fig. 4.1

- (i) Explain what is meant by *electromotive force* (e.m.f.) of 2.0 V.

An e.m.f. of 2.0 V is when 2.0 J of electrical energy converted from other forms of energy to drive 1 C of charge around a complete circuit. [B1]

[1]

- (ii) Calculate the current in the 1.0 Ω resistor and wire XY when switch S_1 is closed.

$$I = \frac{\varepsilon}{1.8 + 1.0 + R_{XY}} = \frac{2.0}{1.8 + 1.0 + 5.0} = 0.26 \text{ (0.256) A [A1]}$$

current = A [1]

- (b) A second circuit which contains Cell B which has an e.m.f. of 1.2 V and negligible resistance is connected to the circuit in (a) at Q and P.

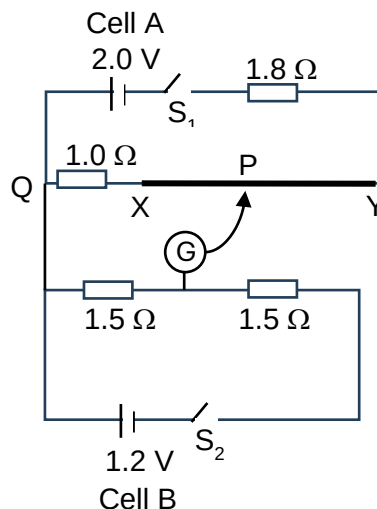


Fig. 4.2

- (i) Calculate the balance length XP required to produce a zero deflection on the galvanometer when switches S_1 and S_2 are both closed..

For zero deflection, the p.d. across QP is 0.60 V. [B1]

$$0.60 = 0.26 (1.0 + R_{XP})$$

$$R_{XP} = 1.3(1.34) \, \Omega \text{ [B1]}$$

$$L_{XP} = \frac{1.3}{5.0} \times 1.000 = 0.26 \text{ m (0.269)[A1]}$$

length = m [3]

- (ii) Describe and explain how the balance length XP will be different if the 1 Ω resistor is replaced with a 1 k Ω resistor.

The pd across the 1 k Ω , $V_{1k\Omega} = \frac{1000}{1000 + 1.8 + 5} \times 2.0 = 1.99 \text{ V OR}$
 The pd across the 1 k Ω is almost 2V or much greater than 0.60 V. [B1]
 Null deflection is not possible. [B1]

[2]

- (iii) if both 1.5 Ω resistors are replaced with identical LDRs, describe and explain how the balance length XP will be change as the light intensity in the laboratory decreases.

The two LDR have the same resistance even as the light intensity change.
 The p.d. across QP is remains 0.60 V. [B1]
 The XP is also unchanged. [B1]

[2]

[Total: 9]